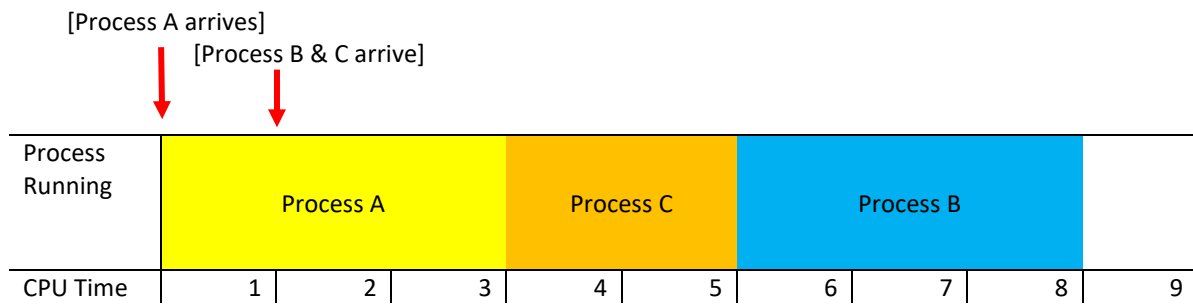


Quiz 2 – Applied OS: Cyber Defense & AI Agents

L2: Student Name: _____

Q1			Q2		Total
(2)	(2)	(2)	(2)	(2)	(10)

Q1 (6)



a) What is the average turnaround time? (2)

Turnaround Time (TT) = Completion Time - Arrival Time.

- Process A $TT_A = 3 - 0 = 3$
- Process C $TT_C = 5 - 1 = 4$
- Process B $TT_B = 8 - 1 = 7$
- Total $TT = 3 + 4 + 7 = 14$
- $TT_{ave.} = \frac{14}{3} = 4\frac{2}{3} = 4.666$ time units.

(b) What is the average response time? (2)

Response Time (RT) is calculated as First Run Time - Arrival Time.

- Process A: $RT_A: 0 - 0 = 0$
- Process C: $RT_C: 3 - 1 = 2$
- Process B: $RT_B: 5 - 1 = 4$
- Total $RT = 0 + 2 + 4 = 6$
- $RT_{ave.} = \frac{6}{3} = 2$ time units.

(c) What type of scheduling policy is this? (2)

STJ and STCF both are correct.

Q2

(4)

(a) What is the `<unistd.h>` library for?

(2)

The `<unistd.h>` (Unix Standard) header file is the C/C++ library that provides access to the POSIX operating system API. It is required to invoke foundational system calls, including process management functions (e.g., `fork`, `exec`, `getpid`) and basic file system I/O functions (e.g., `read`, `write`, `close`).

(b) In terms of PID, what is the difference between `fork()` and `exec()`?

(2)

- `fork()`: Spawns an entirely new child process, hence a new unique Process ID (PID).
- `exec()`: Replaces the current process's memory space and executable code with a new program, hence, retains its original PID.